

Example 2

Find the fundamental matrix of the system $\mathbf{x}' = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \mathbf{x}$.

1– e^{At} , Direct Method

Solution: Since $\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2 = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, and these two matrices commute, we have:

$$\begin{aligned} \exp(\mathbf{A}t) &= \exp\left(\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} t\right) \cdot \exp\left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} t\right) \\ &= \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{2t} \end{bmatrix} \left(\mathbf{I} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} t + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}^2 \frac{t^2}{2!} + \cdots \right). \end{aligned}$$

However, $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, so the series only has two terms. Thus, the fundamental matrix is:

$$\exp(\mathbf{A}t) = e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

2– e^{At} , Laplace Inverse Method

Solution: Since $\mathbf{A} = \mathbf{A}_1 + \mathbf{A}_2 = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, we have:

$$e^{\mathbf{A}t} = e^{(\mathbf{A}_1 + \mathbf{A}_2)t} = e^{\mathbf{A}_1 t} \cdot e^{\mathbf{A}_2 t}$$

First,

$$e^{\mathbf{A}_1 t} = \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{2t} \end{bmatrix}$$

To find $e^{\mathbf{A}_2 t}$, we have

$$e^{\mathbf{A}_2 t} = \mathcal{L}^{-1} \{ (s\mathbf{I} - \mathbf{A}_2)^{-1} \}, \quad \mathbf{A}_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

where $\mathcal{L}^{-1}$ denotes the inverse Laplace transform.

$$s\mathbf{I} - \mathbf{A}_2 = s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 0 & s \end{bmatrix}$$

$$(s\mathbf{I} - \mathbf{A}_2)^{-1} = \frac{1}{s^2} \begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s^2} \\ 0 & \frac{1}{s} \end{bmatrix}$$

$$\mathcal{L}^{-1} \{ (s\mathbf{I} - \mathbf{A}_2)^{-1} \} = \begin{bmatrix} \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} & \mathcal{L}^{-1} \left\{ \frac{1}{s^2} \right\} \\ \mathcal{L}^{-1} \{ 0 \} & \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} \end{bmatrix} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

$$\implies e^{\mathbf{A}_2 t} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

Finally,

$$e^{\mathbf{A} t} = \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{2t} \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

3– Eigenvalues Method

Find eigenvalues

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \det \left(\begin{bmatrix} 2 - \lambda & 1 \\ 0 & 2 - \lambda \end{bmatrix} \right) = (2 - \lambda)^2 = 0 \implies \lambda_1 = \lambda_2 = 2$$

Find eigenvector(s)

$$(\mathbf{A} - 2\mathbf{I})\mathbf{K}_1 = \mathbf{0} \implies \mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

Since we only found one eigenvector, we should find $\mathbf{P}$

$$(\mathbf{A} - 2\mathbf{I})\mathbf{P} = \mathbf{K}_1 \implies \mathbf{P} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

By

$$\mathbf{K}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

fundamental solutions are

$$\mathbf{x}_1(t) = e^{2t} \mathbf{K}_1 = e^{2t} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\mathbf{x}_2(t) = e^{2t} (t\mathbf{K}_1 + \mathbf{P}) = e^{2t} \begin{bmatrix} t \\ 1 \end{bmatrix}$$

Finally, fundamental Matrix ($\Phi(t)$):

$$\Phi(t) = \begin{bmatrix} e^{2t} & te^{2t} \\ 0 & e^{2t} \end{bmatrix} = e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

Example 3 & 5

Find the eigenvalues and corresponding eigenvectors of the matrix $\mathbf{A} = \begin{bmatrix} 3 & 5 \\ -5 & 3 \end{bmatrix}$.

Eigenvalues:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 3 - \lambda & 5 \\ -5 & 3 - \lambda \end{vmatrix} = (3 - \lambda)^2 + 25 = 0$$

$$\implies \lambda_1 = 3 + 5i, \quad \lambda_2 = 3 - 5i$$

Eigenvectors: For $\lambda_1 = 3 + 5i$:

$$\mathbf{A} - (3 + 5i)\mathbf{I} = \begin{bmatrix} -5i & 5 \\ -5 & -5i \end{bmatrix} \implies -ik_1 + k_2 = 0 \implies \mathbf{K}_1 = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

General solution:

$$\mathbf{X}(t) = c_1 \mathbf{K}_1 e^{\lambda_1 t} + c_2 \overline{\mathbf{K}_1} e^{\overline{\lambda_1} t} = c_1 \begin{bmatrix} 1 \\ i \end{bmatrix} e^{(3+5i)t} + c_2 \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{(3-5i)t}$$

where c_1, c_2 are arbitrary constants.

Fundamental matrix

$$\Phi(t) = \begin{bmatrix} \mathbf{K}_1 e^{\lambda_1 t} & \overline{\mathbf{K}_1} e^{\overline{\lambda_1} t} \end{bmatrix} = \begin{bmatrix} e^{(3+5i)t} & e^{(3-5i)t} \\ ie^{(3+5i)t} & -ie^{(3-5i)t} \end{bmatrix}$$

Example 4

Find the eigenvalues and corresponding eigenvectors of the matrix $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$.

Eigenvalue(s):

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 2 - \lambda & 1 \\ -1 & 4 - \lambda \end{vmatrix} = (\lambda - 3)^2 = 0 \implies \lambda = 3 \text{ (repeated)}$$

Eigenvector ($\mathbf{K}$): Solve $(\mathbf{A} - 3\mathbf{I})\mathbf{K} = \mathbf{0}$:

$$\mathbf{A} - 3\mathbf{I} = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \implies -k_1 + k_2 = 0 \implies \mathbf{K} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Generalized Eigenvector ($\mathbf{P}$): Solve $(\mathbf{A} - 3\mathbf{I})\mathbf{P} = \mathbf{K}$:

$$\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \mathbf{P} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \implies -p_1 + p_2 = 1 \implies \mathbf{P} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Fundamental Solutions:

$$\mathbf{x}_1(t) = e^{3t} \mathbf{K} = e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\mathbf{x}_2(t) = e^{3t} (t\mathbf{K} + \mathbf{P}) = e^{3t} \begin{bmatrix} t \\ 1 + t \end{bmatrix}$$

Fundamental Matrix ($\Phi(t)$):

$$\Phi(t) = \begin{bmatrix} e^{3t} & te^{3t} \\ e^{3t} & (1+t)e^{3t} \end{bmatrix} = e^{3t} \begin{bmatrix} 1 & t \\ 1 & 1+t \end{bmatrix}$$

Example 9

Solve the system of linear differential equations

$$\begin{cases} x'_1 = 3x_1 - x_2 + x_3, \\ x'_2 = 2x_1 + x_3, \\ x'_3 = x_1 - x_2 + 2x_3, \end{cases}$$

Solution

First we write the system in a matrix form $\mathbf{x}' = \mathbf{A}\mathbf{x}$ The coefficient matrix is:

$$\mathbf{A} = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}$$

Step 1: Find Eigenvalues

First, construct $\lambda\mathbf{I} - \mathbf{A}$:

$$\lambda\mathbf{I} - \mathbf{A} = \begin{bmatrix} \lambda - 3 & 1 & -1 \\ -2 & \lambda & -1 \\ -1 & 1 & \lambda - 2 \end{bmatrix}$$

Then,

$$\begin{aligned} \det(\lambda\mathbf{I} - \mathbf{A}) &= (\lambda - 3) \det \begin{bmatrix} \lambda & -1 \\ 1 & \lambda - 2 \end{bmatrix} - 1 \det \begin{bmatrix} -2 & -1 \\ -1 & \lambda - 2 \end{bmatrix} - 1 \det \begin{bmatrix} -2 & \lambda \\ -1 & 1 \end{bmatrix} \\ &= (\lambda - 3)(\lambda^2 - 2\lambda + 1) - 1(-2\lambda + 3) - 1(\lambda - 2) \\ &= (\lambda - 3)(\lambda - 1)^2 + 2\lambda - 3 - \lambda + 2 \\ &= \lambda^3 - 5\lambda^2 + 8\lambda - 4 \\ &= (\lambda - 2)^2(\lambda - 1) = 0 \end{aligned}$$

Thus, the eigenvalues are:

- $\lambda_1 = 2$ (algebraic multiplicity 2, repeated twice)
- $\lambda_2 = 1$ (algebraic multiplicity 1, simple eigenvalue)

Step 2: Find Eigenvectors

$$\lambda_1 = 2 : \quad (\mathbf{A} - 2\mathbf{I})\mathbf{K}_1 = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -2 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = 0$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 2 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{array} \right] \xrightarrow{\text{Row Operations}} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

So,

$$\mathbf{K}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Note: It gives only one eigenvector. So, find $\mathbf{P}$.

To find $\mathbf{P}$, solve $(\mathbf{A} - 2\mathbf{I})\mathbf{P} = \mathbf{K}_1$

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & -2 & 1 \\ 1 & -1 & 0 \end{bmatrix} \mathbf{P} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \implies \mathbf{P} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

because,

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 2 & -2 & 1 & 1 \\ 1 & -1 & 0 & 0 \end{array} \right] \xrightarrow{\text{Row Operations}} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Now

$$\lambda_2 = 1 : \quad (\mathbf{A} - 1\mathbf{I})\mathbf{K}_2 = \mathbf{0} \implies \begin{bmatrix} 2 & -1 & 1 \\ 2 & -1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \mathbf{0}$$

$$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 2 & -1 & 1 & 0 \\ 1 & -1 & 1 & 0 \end{array} \right] \xrightarrow{\text{Row Operations}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\implies \mathbf{K}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Fundamental Solutions are

$$\mathbf{x}_1(t) = \mathbf{K}_1 e^{\lambda_1 t} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

$$\mathbf{x}_2(t) = (t\mathbf{K}_1 + \mathbf{P}) e^{\lambda_1 t} = \left(t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) e^{2t} = \begin{bmatrix} t \\ t \\ 1 \end{bmatrix} e^{2t}$$

$$\mathbf{x}_3(t) = \mathbf{K}_2 e^{\lambda_2 t} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t$$

General Solution:

$$\mathbf{x}(t) = c_1 e^{2t} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} t \\ t \\ 1 \end{bmatrix} + c_3 e^t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Fundamental Matrix:

$$\Phi = \begin{bmatrix} (1+t)e^{2t} & -te^{2t} & te^{2t} \\ -e^t + (1+t)e^{2t} & e^t - te^{2t} & te^{2t} \\ -e^t + e^{2t} & e^t - e^{2t} & e^{2t} \end{bmatrix}$$

Example 10

Let

$$\mathbf{A} = \begin{bmatrix} 3 & 5 \\ -5 & 3 \end{bmatrix}, \quad \mathbf{f}(t) = \begin{bmatrix} e^{-t} \\ 0 \end{bmatrix}.$$

Find the solution $\mathbf{X}(t)$ of the equation $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f}(t)$ that satisfies the initial condition $\mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ where fundamental matrix is

$$\Phi = e^{3t} \begin{bmatrix} \cos 5t & \sin 5t \\ -\sin 5t & \cos 5t \end{bmatrix}$$

Solution The general solution of this nonhomogenous system is

$$\mathbf{x}(t) = \Phi(t)\mathbf{c} + \Phi(t) \int_0^t \Phi^{-1}(s) \begin{bmatrix} e^{-s} \\ 0 \end{bmatrix} ds,$$

and since $\mathbf{x}(0)$ is given

$$\mathbf{c} = \Phi^{-1}(0)\mathbf{x}(0)$$

Here,

$$\Phi^{-1}(s) = e^{-3s} \begin{bmatrix} \cos 5s & -\sin 5s \\ \sin 5s & \cos 5s \end{bmatrix}$$

$$\Phi^{-1}(s)\mathbf{f}(s) = e^{-3s} \begin{bmatrix} \cos 5s & -\sin 5s \\ \sin 5s & \cos 5s \end{bmatrix} \begin{bmatrix} e^{-s} \\ 0 \end{bmatrix} = e^{-4s} \begin{bmatrix} \cos 5s \\ \sin 5s \end{bmatrix}$$

Find the integral

$$\int_0^t e^{-4s} \begin{bmatrix} \cos 5s \\ \sin 5s \end{bmatrix} ds = \frac{1}{41} \begin{bmatrix} e^{-4t}(-4 \cos 5t + 5 \sin 5t) + 4 \\ e^{-4t}(-4 \sin 5t - 5 \cos 5t) + 5 \end{bmatrix}$$

Then,

$$\begin{aligned} \Phi(t) \int_0^t \Phi^{-1}(s)\mathbf{f}(s)ds &= \frac{e^{3t}}{41} \begin{bmatrix} \cos 5t & \sin 5t \\ -\sin 5t & \cos 5t \end{bmatrix} \begin{bmatrix} e^{-4t}(-4 \cos 5t + 5 \sin 5t) + 4 \\ e^{-4t}(-4 \sin 5t - 5 \cos 5t) + 5 \end{bmatrix} \\ &= \frac{e^{3t}}{41} \begin{bmatrix} -4e^{-4t} + 4 \cos 5t + 5 \sin 5t \\ -5e^{-4t} - 4 \sin 5t + 5 \cos 5t \end{bmatrix} \end{aligned}$$

Now, we find $\mathbf{c}$

$$\mathbf{c} = \Phi^{-1}(0)\mathbf{x}(0) = \mathbf{x}(0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

By replacing them all in

$$\mathbf{x}(t) = \Phi(t)\mathbf{c} + \Phi(t) \int_0^t \Phi^{-1}(s)\mathbf{f}(s)ds$$

we finally get,

$$\begin{aligned} \mathbf{x}(t) &= e^{3t} \begin{bmatrix} \cos 5t & \sin 5t \\ -\sin 5t & \cos 5t \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \frac{e^{3t}}{41} \begin{bmatrix} -4e^{-4t} + 4 \cos 5t + 5 \sin 5t \\ -5e^{-4t} - 4 \sin 5t + 5 \cos 5t \end{bmatrix} \\ &= \frac{e^{3t}}{41} \begin{bmatrix} 4 \cos 5t + 46 \sin 5t \\ 46 \cos 5t - 4 \sin 5t \end{bmatrix} - \frac{e^{-t}}{41} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \end{aligned}$$

Notes and Clarifications

1 – The general solution of homogenous system

$$\mathbf{x}_h(t) = \mathbf{\Phi} \mathbf{c} = e^{3t} \begin{bmatrix} \cos 5t & \sin 5t \\ -\sin 5t & \cos 5t \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

2 – The particular solution obtained by variation of parameter is

$$\mathbf{x}_p(t) = \mathbf{\Phi}(t) \int_0^t \mathbf{\Phi}^{-1}(s) \mathbf{f}(s) ds = \frac{e^{3t}}{41} \begin{bmatrix} -4e^{-4t} + 4 \cos 5t + 5 \sin 5t \\ -5e^{-4t} - 4 \sin 5t + 5 \cos 5t \end{bmatrix}$$

3 – The general solution of non-homogenous system

$$\begin{aligned} \mathbf{x}(t) &= \mathbf{x}_h(t) + \mathbf{x}_p(t) \\ &= e^{3t} \begin{bmatrix} \cos 5t & \sin 5t \\ -\sin 5t & \cos 5t \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \frac{e^{3t}}{41} \begin{bmatrix} -4e^{-4t} + 4 \cos 5t + 5 \sin 5t \\ -5e^{-4t} - 4 \sin 5t + 5 \cos 5t \end{bmatrix} \end{aligned}$$

4 – The solution of IVP: As we found $\mathbf{c} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\begin{aligned} \mathbf{x}(t) &= e^{3t} \begin{bmatrix} \cos 5t & \sin 5t \\ -\sin 5t & \cos 5t \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \frac{e^{3t}}{41} \begin{bmatrix} -4e^{-4t} + 4 \cos 5t + 5 \sin 5t \\ -5e^{-4t} - 4 \sin 5t + 5 \cos 5t \end{bmatrix} \\ &= \frac{e^{3t}}{41} \begin{bmatrix} 4 \cos 5t + 46 \sin 5t \\ 46 \cos 5t - 4 \sin 5t \end{bmatrix} - \frac{e^{-t}}{41} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \end{aligned}$$

Some Important Remarks

Remark 1– Fundamental matrix of a system, $\Phi(t)$, is not unique.

In example 1, $\mathbf{x}' = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \mathbf{x}$, we found

$$\Phi(t) = \begin{bmatrix} e^{2t} & te^{2t} \\ 0 & e^{2t} \end{bmatrix}$$

We can show that $\begin{bmatrix} te^{2t} & e^{2t} \\ e^{2t} & 0 \end{bmatrix}$ is a fundamental matrix as well, because

- The columns still satisfy $\mathbf{x}' = \mathbf{A}\mathbf{x}$
- The determinant condition is provided

$$\det \left(\begin{bmatrix} te^{2t} & e^{2t} \\ e^{2t} & 0 \end{bmatrix} \right) \neq 0$$

Remark 2– For matrix $\mathbf{A}$, the matrix $e^{\mathbf{A}t}$ is unique.

Remark 3– The matrix $e^{\mathbf{A}t}$ is a fundamental matrix of $\mathbf{x}' = \mathbf{A}\mathbf{x}$.

Note that the matrix $e^{\mathbf{A}t}$ is a fundamental matrix of system $\mathbf{x}' = \mathbf{A}\mathbf{x}$, but a fundamental matrix of this system, is not necessarily equal to $e^{\mathbf{A}t}$

Remark 4– For general solution of $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f}(t)$

$$\mathbf{x}(t) = \Phi(t)\mathbf{c} + \Phi(t) \int_0^t \Phi^{-1}(s)\mathbf{f}(s) ds$$

However, if we have $e^{\mathbf{A}t}$,

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{c} + e^{\mathbf{A}t} \int_0^t e^{-\mathbf{A}s}\mathbf{f}(s) ds$$

Remark 5 – In the above formulas

– For $\Phi(t) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$,

$$\Phi^{-1}(t) = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

– If $\Phi(t) = e^{\mathbf{A}t}$, the inverse simplifies to $e^{-\mathbf{A}t}$.

Remark 6 – For system $\mathbf{x}' = \mathbf{A}\mathbf{x}$, if $\Phi(t)$ is a fundamental matrix then

$$e^{\mathbf{A}t} = \Phi(t)\Phi^{-1}(0),$$

Exercises 5.2

3. Calculate the eigenvalues and corresponding eigenvectors of the following matrices:

$$(1) \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \quad (2) \begin{bmatrix} 2 & -3 & 3 \\ 4 & -5 & 3 \\ 4 & -4 & 2 \end{bmatrix}$$

$$(3) \begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 0 & 1 \end{bmatrix} \quad (4) \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$$

$$(5) \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \quad (6) \begin{bmatrix} 3 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

Solution

Solution for Matrix (1): $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$

Step 1: Find eigenvalues

$$|\lambda \mathbf{I} - \mathbf{A}| = \begin{vmatrix} \lambda - 1 & -2 \\ -4 & \lambda - 3 \end{vmatrix} = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0$$

so, eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = -1$

Step 2: Compute Eigenvectors

For $\lambda_1 = 5$:

$$(5\mathbf{I} - \mathbf{A})\mathbf{K}_1 = \begin{bmatrix} 4 & -2 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = 0 \implies \mathbf{K}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

For $\lambda_2 = -1$:

$$(-\mathbf{I} - \mathbf{A})\mathbf{K}_2 = \begin{bmatrix} -2 & -2 \\ -4 & -4 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = 0 \implies \mathbf{K}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

4. Find a fundamental matrix for the system $\mathbf{x}' = \mathbf{A}\mathbf{x}$, and compute $e^{\mathbf{A}t}$, where $\mathbf{A}$ is:

$$(1) \begin{bmatrix} -2 & 1 \\ -1 & 2 \end{bmatrix} \quad (2) \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$$

$$(3) \begin{bmatrix} 2 & -3 & 3 \\ 4 & -5 & 3 \\ 4 & -4 & 2 \end{bmatrix} \quad (4) \begin{bmatrix} 1 & 0 & 3 \\ 8 & 1 & -1 \\ 5 & 1 & -1 \end{bmatrix}$$

Solution

Solution for Matrix (1): $\mathbf{A} = \begin{bmatrix} -2 & 1 \\ -1 & 2 \end{bmatrix}$

Step 1: Find Eigenvalues

$$|\lambda \mathbf{I} - \mathbf{A}| = \begin{vmatrix} \lambda + 2 & -1 \\ 1 & \lambda - 2 \end{vmatrix} = (\lambda + 2)(\lambda - 2) + 1 = \lambda^2 - 3$$

So, eigenvalues are $\lambda_1 = \sqrt{3}$, $\lambda_2 = -\sqrt{3}$.

Step 2: Find Eigenvectors

- For $\lambda_1 = \sqrt{3}$:

$$(\sqrt{3}\mathbf{I} - \mathbf{A})\mathbf{K}_1 = \begin{bmatrix} \sqrt{3} + 2 & -1 \\ 1 & \sqrt{3} - 2 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = 0 \implies \mathbf{K}_1 = \begin{bmatrix} 1 \\ \sqrt{3} + 2 \end{bmatrix}$$

From the first row, $(\sqrt{3} + 2)x_1 - x_2 = 0$, let $x_1 = 1$.

- For $\lambda_2 = -\sqrt{3}$:

$$(-\sqrt{3}\mathbf{I} - \mathbf{A})\mathbf{K}_2 = \begin{bmatrix} -\sqrt{3} + 2 & -1 \\ 1 & -\sqrt{3} - 2 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = 0 \implies \mathbf{K}_2 = \begin{bmatrix} 1 \\ 2 - \sqrt{3} \end{bmatrix}$$

From the first row $(2 - \sqrt{3})x_1 - x_2 = 0$, let $x_1 = 1$.

Step 3: Fundamental Matrix

$$\Phi(t) = \begin{bmatrix} e^{\sqrt{3}t} & e^{-\sqrt{3}t} \\ (\sqrt{3} + 2)e^{\sqrt{3}t} & (2 - \sqrt{3})e^{-\sqrt{3}t} \end{bmatrix}$$

Step 4: Compute $e^{\mathbf{A}t}$

Note that

$$e^{\mathbf{A}t} = \mathbf{\Phi}(t)\mathbf{\Phi}^{-1}(0)$$

Therefore,

- First, $\mathbf{\Phi}(0) = \begin{bmatrix} 1 & 1 \\ \sqrt{3} + 2 & 2 - \sqrt{3} \end{bmatrix}$

- Find its inverse $\mathbf{\Phi}^{-1}(0) = \frac{1}{-2\sqrt{3}} \begin{bmatrix} 2 - \sqrt{3} & -1 \\ -\sqrt{3} - 2 & 1 \end{bmatrix}$

- Multiply $\mathbf{\Phi}(t)\mathbf{\Phi}^{-1}(0)$:

$$e^{\mathbf{A}t} = \frac{1}{2\sqrt{3}} \begin{bmatrix} (\sqrt{3} + 2)e^{\sqrt{3}t} + (2 - \sqrt{3})e^{-\sqrt{3}t} & e^{\sqrt{3}t} - e^{-\sqrt{3}t} \\ e^{\sqrt{3}t} - e^{-\sqrt{3}t} & (2 - \sqrt{3})e^{\sqrt{3}t} + (\sqrt{3} + 2)e^{-\sqrt{3}t} \end{bmatrix}$$

Solution

Solution for Matrix (2) $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$:

Step 1: Find Eigenvalues

$$|\lambda \mathbf{I} - \mathbf{A}| = \begin{vmatrix} \lambda - 1 & -2 \\ -4 & \lambda - 3 \end{vmatrix} = (\lambda - 1)(\lambda - 3) - 8 = \lambda^2 - 4\lambda - 5 = 0$$

So, eigenvalues are $\lambda_1 = 5$, $\lambda_2 = -1$.

Step 2: Find Eigenvectors

- For $\lambda_1 = 5$:

$$(5\mathbf{I} - \mathbf{A})\mathbf{K}_1 = \begin{bmatrix} 4 & -2 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = 0 \implies \mathbf{K}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- For $\lambda_2 = -1$:

$$(-\mathbf{I} - \mathbf{A})\mathbf{K}_1 = \begin{bmatrix} -2 & -2 \\ -4 & -4 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = 0 \implies \mathbf{K}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Step 3: Fundamental Matrix $\Phi(t)$

$$\Phi(t) = \begin{bmatrix} e^{5t} & e^{-t} \\ 2e^{5t} & -e^{-t} \end{bmatrix}$$

Step 5: Compute e^{At}

$$\Phi(0) = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \implies \Phi^{-1}(0) = \frac{1}{-3} \begin{bmatrix} -1 & -1 \\ -2 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$$

Finally,

$$\begin{aligned} e^{At} &= \Phi(t)\Phi^{-1}(0) = \frac{1}{3} \begin{bmatrix} e^{5t} & e^{-t} \\ 2e^{5t} & -e^{-t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} e^{5t} + 2e^{-t} & e^{5t} - e^{-t} \\ 2e^{5t} - 2e^{-t} & 2e^{5t} + e^{-t} \end{bmatrix} \end{aligned}$$

5. Find a fundamental matrix for the system $\mathbf{x}' = \mathbf{A}\mathbf{x}$, and find the solution $\varphi(t)$ satisfying the initial condition $\varphi(0) = \boldsymbol{\eta}$:

$$(1) \mathbf{A} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}, \boldsymbol{\eta} = \begin{bmatrix} 3 \\ 3 \end{bmatrix};$$

$$(2) \mathbf{A} = \begin{bmatrix} 1 & 0 & 3 \\ 8 & 1 & -1 \\ 5 & 1 & -1 \end{bmatrix}, \boldsymbol{\eta} = \begin{bmatrix} 0 \\ -2 \\ -7 \end{bmatrix};$$

$$(3) \mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 0 & 1 \end{bmatrix}, \boldsymbol{\eta} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Solution

(1) Solution for Matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ and $\boldsymbol{\eta} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$:

Step 1: Find Eigenvalues

$$|\lambda \mathbf{I} - \mathbf{A}| = \lambda^2 - 4\lambda - 5 = 0 \implies \lambda_1 = 5, \lambda_2 = -1$$

Step 2: Find Eigenvectors

- For $\lambda_1 = 5$: $\mathbf{K}_1 = k \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

- For $\lambda_2 = -1$: $\mathbf{K}_2 = k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Step 3: Fundamental Matrix $\Phi(t)$

$$\Phi(t) = \begin{bmatrix} e^{5t} & e^{-t} \\ 2e^{5t} & -e^{-t} \end{bmatrix}$$

Step 4: The general solution of this system

$$\mathbf{X}(t) = \Phi(t)\mathbf{c}$$

Step 5: Compute $\mathbf{c}$:

$$\mathbf{X}(0) = \Phi(0)\mathbf{c} \implies \mathbf{c} = \Phi^{-1}(0)\boldsymbol{\eta}, \quad \text{where} \quad \boldsymbol{\eta} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\Phi(0) = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \implies \Phi^{-1}(0) = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$$

$$\mathbf{c} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Step 6: Final Solution

$$\boldsymbol{\varphi}(t) = \Phi(t)\mathbf{c} = \begin{bmatrix} 2e^{5t} + e^{-t} \\ 4e^{5t} - e^{-t} \end{bmatrix}$$

6 – Find the solution of $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$ where

(1) $\mathbf{x}(0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$, $\mathbf{f}(t) = \begin{bmatrix} e^t \\ 1 \end{bmatrix}$

(2) $\mathbf{x}(0) = \mathbf{0}$, $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$, $\mathbf{f}(t) = \begin{bmatrix} 0 \\ 0 \\ e^{-t} \end{bmatrix}$

$$(3) \quad \mathbf{x}(0) = \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix}, \quad A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}, \quad \mathbf{f}(t) = \begin{bmatrix} \sin t \\ -2 \cos t \end{bmatrix}$$

Solution

Solution for (1)

$$\mathbf{x}(0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}, \quad \mathbf{f}(t) = \begin{bmatrix} e^t \\ 1 \end{bmatrix}$$

The solution uses the **variation of parameters**:

$$\mathbf{x}(t) = \Phi(t)\mathbf{c} + \Phi(t) \int_0^t \Phi^{-1}(s)\mathbf{f}(s) ds$$

where $\mathbf{c} = \Phi^{-1}(0)\boldsymbol{\eta}$

Step 1: Find Eigenvalues/Eigenvectors

$$\det(\lambda I - A) = (\lambda - 5)(\lambda + 1) = 0 \implies \lambda_1 = 5, \lambda_2 = -1$$

Thus,

$$\mathbf{K}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (\text{for } \lambda_1 = 5), \quad \mathbf{K}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (\text{for } \lambda_2 = -1)$$

Step 2: Fundamental matrix

$$\Phi(t) = \begin{bmatrix} e^{5t} & e^{-t} \\ 2e^{5t} & -e^{-t} \end{bmatrix}$$

Step 3: Compute the formula

$$\mathbf{x}(t) = \Phi(t)\mathbf{c} + \Phi(t) \int_0^t \Phi^{-1}(s)\mathbf{f}(s) ds \quad \text{where} \quad \mathbf{c} = \Phi^{-1}(0)\boldsymbol{\eta}$$

$$\Phi(t) = \begin{bmatrix} e^{5t} & e^{-t} \\ 2e^{5t} & -e^{-t} \end{bmatrix} \implies \Phi^{-1}(t) = \frac{1}{3} \begin{bmatrix} e^{-5t} & e^{-5t} \\ 2e^t & -e^t \end{bmatrix}$$

$$\Phi^{-1}(s)\mathbf{f}(s) = \frac{1}{3} \begin{bmatrix} e^{-4s} + e^{-5s} \\ 2e^{2s} - e^s \end{bmatrix}$$

$$\int_0^t \Phi^{-1}(s) \mathbf{f}(s) ds = \frac{1}{3} \begin{bmatrix} \frac{9}{20} - \frac{1}{4}e^{-4t} - \frac{1}{5}e^{-5t} \\ e^{2t} - e^t \end{bmatrix}$$

$$\Phi(t) \cdot \int_0^t \Phi^{-1}(s) \mathbf{f}(s) ds = \begin{bmatrix} \frac{3}{20}e^{5t} + \frac{1}{4}e^t - \frac{2}{5} \\ \frac{3}{10}e^{5t} - \frac{1}{2}e^t + \frac{1}{5} \end{bmatrix}$$

Also

$$\mathbf{c} = \Phi^{-1}(0)\boldsymbol{\eta} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\Phi(t)\mathbf{c} = \begin{bmatrix} -e^{-t} \\ e^{-t} \end{bmatrix}$$

Finally,

$$\mathbf{x}(t) = \begin{bmatrix} \frac{3}{20}e^{5t} + \frac{1}{4}e^t - e^{-t} - \frac{2}{5} \\ \frac{3}{10}e^{5t} - \frac{1}{2}e^t + e^{-t} + \frac{1}{5} \end{bmatrix}$$

Example – Find the solution of $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$ where

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{f}(t) = \begin{bmatrix} e^t \\ 0 \end{bmatrix}$$

and $\mathbf{x}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, also the exponential matrix $\mathbf{A}$ is given as

$$e^{At} = e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

Solution The general solution is:

$$\mathbf{x}(t) = e^{At}\mathbf{c} + e^{At} \int_0^t e^{-As} \mathbf{f}(s) ds$$

where $\mathbf{c} = \mathbf{x}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$e^{\mathbf{A}t} = e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \implies e^{-\mathbf{A}t} = e^{-2t} \begin{bmatrix} 1 & -t \\ 0 & 1 \end{bmatrix}$$

$$e^{-\mathbf{A}s} \mathbf{f}(s) = e^{-2s} \begin{bmatrix} 1 & -s \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^s \\ 0 \end{bmatrix} = e^{-2s} \begin{bmatrix} e^s \\ 0 \end{bmatrix} = \begin{bmatrix} e^{-s} \\ 0 \end{bmatrix}$$

$$\int_0^t e^{-\mathbf{A}s} \mathbf{f}(s) ds = \int_0^t \begin{bmatrix} e^{-s} \\ 0 \end{bmatrix} ds = \begin{bmatrix} 1 - e^{-t} \\ 0 \end{bmatrix}$$

$$e^{\mathbf{A}t} \int_0^t e^{-\mathbf{A}s} \mathbf{f}(s) ds = e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 - e^{-t} \\ 0 \end{bmatrix} = e^{2t} \begin{bmatrix} 1 - e^{-t} \\ 0 \end{bmatrix} = \begin{bmatrix} e^{2t} - e^t \\ 0 \end{bmatrix}$$

Also we have

$$e^{\mathbf{A}t} \mathbf{c} = e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} (1-t)e^{2t} \\ -e^{2t} \end{bmatrix}$$

Now we replace them in the formula

$$\mathbf{x}(t) = \begin{bmatrix} (1-t)e^{2t} \\ -e^{2t} \end{bmatrix} + \begin{bmatrix} e^{2t} - e^t \\ 0 \end{bmatrix} = \begin{bmatrix} (2-t)e^{2t} - e^t \\ -e^{2t} \end{bmatrix}$$

Example— Find the fundamental matrix of system $\mathbf{x}' = \mathbf{A}\mathbf{x}$ for the following matrices:

$$(1) \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} \quad (2) \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 3 \\ 0 & -1 & 2 \end{bmatrix}$$

$$(3) \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (4) \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(5) \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (6) \begin{bmatrix} 3 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

Solution The solution of (6) $\mathbf{A} = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & -1 \end{bmatrix}$

Step1. Eigenvalues are 3, 3, -1

Step 2. Eigenvectors

For $\lambda_1 = 3$ (repeated)

$$(\mathbf{A} - 3I)\mathbf{K}_1 = \mathbf{0} \implies \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & -4 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \mathbf{0} \implies \mathbf{K}_1 = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$$

Find $\mathbf{P}$

$$(\mathbf{A} - 3I)\mathbf{P} = \mathbf{K}_1 \implies \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & -4 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} \implies \mathbf{P} = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$$

For $\lambda_2 = -1$:

$$(\mathbf{A} + I)\mathbf{K}_2 = \mathbf{0} \implies \begin{bmatrix} 4 & 0 & 0 \\ 1 & 4 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \mathbf{0} \implies \mathbf{K}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

So,

$$\mathbf{X}_1(t) = \mathbf{K}_1 e^{3t} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} e^{3t},$$

$$\mathbf{X}_2(t) = (\mathbf{K}_1 t + \mathbf{P}) e^{3t} = \left(\begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} t + \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} \right) e^{3t},$$

$$\mathbf{X}_3(t) = \mathbf{K}_2 e^{-t} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t}$$

Step 3. Fundamental matrix is

$$\Phi(t) = \begin{bmatrix} 0 & 4e^{3t} & e^{-t} \\ 4e^{3t} & e^{3t}(1+4t) & 0 \\ e^{3t} & te^{3t} & 0 \end{bmatrix}$$